

Realization of a Modular Reconfigurable Robot for Rough Terrain

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Abstract – Reconfigurable robotic systems are made from a large number of independent modules that can be used to form various robot shapes. As a result, the robots have the capability to adopt optimized configurations to suit different tasks and environments.

The object of this research work is to build a mobile robot for rough terrain. In this paper, we developed a novel modular reconfigurable robotic system called JL-I. JL-I is a distributed system composed of several identical modules. Each module is self-contained including two powered tracks, a docking mechanism, pose-adjusting mechanisms, batteries, communication devices and microprocessors. The docking mechanisms allow the modules to connect or disconnect under computer control, and between each two adjacent modules, the pose-adjusting mechanisms are able to form an active spherical joint, which endows the robot with the abilities of changing shapes in three dimensions. This design of JL-I module realizes versatile robotic motion and configurations. Many useful functionalities such as getting recovery when the robot is turnover, getting over vertical steps can be obtained by different chain of motions. This paper describes the development, hardware and software construction of the JL-I robot. An actual system with three modules was build and basic configurations were examined.

Index Terms – *Modular robots, reconfiguration, active spherical joint, robot architectures.*

I. INTRODUCTION

Reconfigurable robots are composed of a large number of modules. Each of them has the ability to connect to or disconnect from the robot. By rearranging the modules' poses and positions with respect to adjacent modules, the robot can change its structure and functionality according to tasks and environments. For instance, a modular reconfigurable robot could change its shape into a snake to reach into a narrow space during a rescue operation, into a rolling type to pass over flat terrains and split into many smaller segments to perform tasks in parallel [1].

The reconfigurable robotic systems can be classified according to whether or not their modules are organized in a lattice. Lattice-based robots are usually homogeneous, they change shapes by moving into positions on a virtual grid and it may only move to neighbouring positions within the lattice. Compared with lattice robots, non-lattice robots can not only translate while reconfiguring but also separate their reconfiguration and locomotion. This separation enables the

robots to reconfigure and then select a suitable configuration-dependent gait [2]. Therefore, this kind of robots is more suitable for carrying out tasks under different environments.

Recently, the flexibility of reconfigurable robot has attracted many researchers and many systems have been developed [3]-[11]. These robots have many advantages over traditional robots, but they have few configurations due to simple connecting and pose-adjusting mechanisms. They can only deal with relatively smooth terrains. For example, a kind of reconfigurable mobile robot proposed by Myoungkuk Park, Woojin Chung and Munsang Kim in Korea is only able to change the configuration passively [12], a modular reconfigurable robot developed by Wang Tianmiao, Zou Dan in Beijing University of Aeronautics and Astronautics has only 1 DOF in pose-adjusting mechanism [13], and the Millibot developed by H. Benjamin Brown, Jr., J. Michael. in Carnegie Mellon University [14], the Swart-Bot proposed by E. Sahin, T.H.Labella, V.Trianni in Université Libre de Bruxelles[15] have one 1-DOF pose-adjusting mechanism and one coupler to change the configuration with regard to neighboring modules, a shape-shifting robot proposed by Jinguo Liu, Bin Li, et al in Chinese Academy of Sciences has the capability to change the pose relation between two neighbouring modules in 2 DOF, however, its docking systems need external help to implement all the reconfiguration processes [16][17]. Although these reconfigurable mobile robots mentioned above have many good characteristics, such as versatility and robustness, they are not very fit for rough terrain operations for few robot configurations.

This project aims to build a field robot for unstructured environments based on reconfiguration conception. We developed a modular reconfigurable mobile robot called JL-I, which has several modules and each module is self-contained. The JL-I modules can not only link into a train configuration to deal with rugged terrain, to change robot shape in three dimensions according to different tasks, but also disconnect from the robot to perform activities individually.

This paper consists of six sections. The development of the JL-I is introduced in detail in Section II, the hardware and software architecture are described in Section III, reconfigurations and functionalities are given in Section IV. Experimental results are shown in Sections V, and concluding remarks are given in the last section.

II. THE DEVELOPMENT OF THE ROBOT - JL-I

A. Philosophy of design

Fig. 1 shows a schematic view of the JL-I robot. The robot is composed by several identical modules. The module is self-contained including two powered tracks, a docking mechanism and pose-adjusting mechanisms composed by a serial mechanism and a parallel mechanism as shown in Fig. 2. The serial and parallel mechanisms are used to form a 3-DOF active spherical joint for each two adjacent modules. The docking mechanism enables the JL-I module to link with the other modules to form a train configuration for the uneven environment or disconnect from the robot to perform mobile distributed activities. The active spherical joint allows the modules to rotate about the X-axis, Y-axis and Z-axis. By repeating such basic operations, different configurations in 3 dimensions are achieved, and consequently to obtain many useful functionalities.

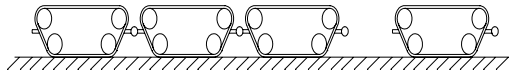


Fig. 1 Schematic view of the JL-I robot.

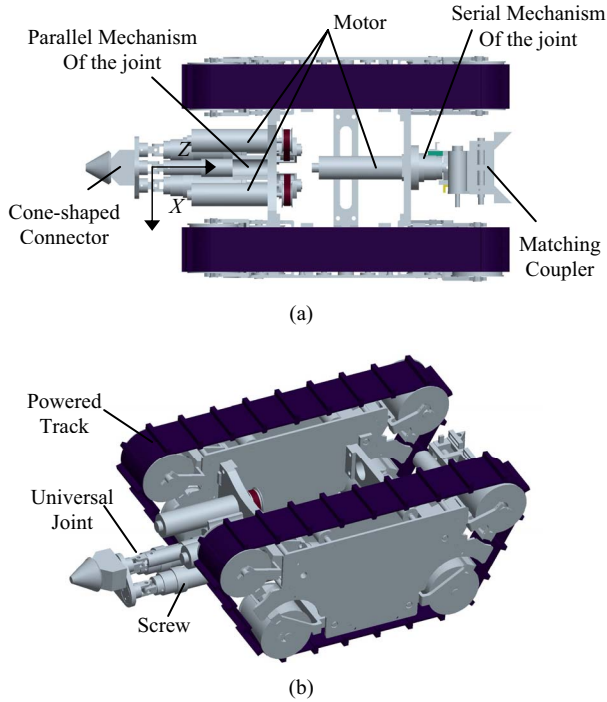


Fig. 2 Schematic view of the proposed module.

B. Performance of the robot

The JL-I robot has good performance for rough terrain. It has many advantages as train-type mobile robots have, such as smoothing the undulating motion of the gravity center of the whole robot, increasing the safety when the robot moves on rugged terrain, highly efficient by executing the work

synchronously [18]. Besides these three advantages, it has two great characteristics.

1) *Highly robust*: If a traditional mobile robot lost its balance, it would be disabled. While the JL-I robot with active spherical joints can repair itself by implementing $90^\circ/180^\circ$ self-righting motion.

2) *Highly adaptive*: JL-I has a large variety of configurations due to the active spherical joints and the docking mechanisms. With all the modules configured appropriately, it is capable of adapting to the rugged terrain, getting over steps, passing over narrow places and etc.

III. HARDWARE AND SOFTWARE REALIZATION

This section illustrates the implementation of the JL-I robot. The discussion presents an over view of the mechanical, electronic and software realization.

A. Mechanical Design

The module has three basic components: two powered tracks, a docking system and an active spherical joint which includes a serial mechanism and a parallel mechanism. The main issues addressed here are the design of the docking system and the 3-DOF joint.

1) *Docking system*. The docking system as shown in Fig. 3 is based on a cone-shaped connector that registers with a matching coupler to constrain 4 DOFs. The latching mechanism in the coupler is two sliders propelled by a motor-driven screw. The sliders and the coupler with a matching funnel have the ability to guide the connector to mate with the cavity, which enable the module to self-align $\pm 30\text{mm}$ lateral offsets and $\pm 45^\circ$ directional offsets. Between each slider and the coupler, there are two mating planes that constrain the rest of DOFs between them. Thus, motions can be transferred after the connector has been locked into the coupler.

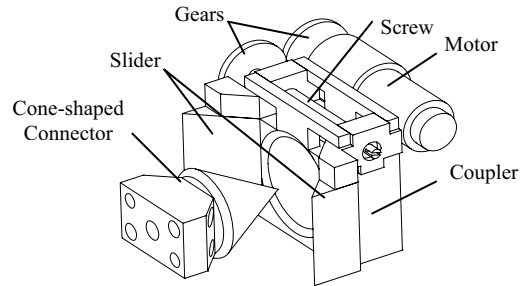


Fig. 3 Mechanical diagram of the docking system.

2) *Active spherical joint*. The robot's reconfiguring motion is performed by the rotation of the active spherical joint which has 3 DOFs. Two of them achieved by the parallel mechanism are rotation about the joint's x and y axis. The third one obtained by the serial mechanism is rotation about the z axis.

The parallel part of the joint is shown in Fig. 4, where each leg consists of a driving platform, a universal joint, a screw, a synchronous belt system, a DC motor and a base platform. The driving platform is also connected to the

knighthead by means of a universal joint. The other end of the knighthead is fixed to the base platform. By revolving the screw, the driving platform can be manipulated with respect to the base platform.

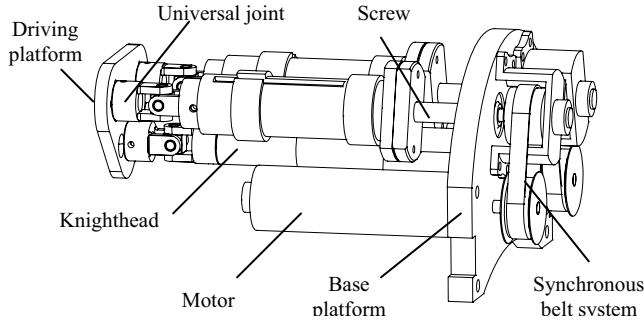


Fig. 4 Mechanical diagram of the parallel part of the active spherical joint.

The joint's serial part is mainly made up of a Minimotor geared motor. This unit provides a continuous torque of 3.5Nm at a speed of 30rpm.

Currently, we have built three modules equipped with all the needed mechanisms as shown in Fig. 5.

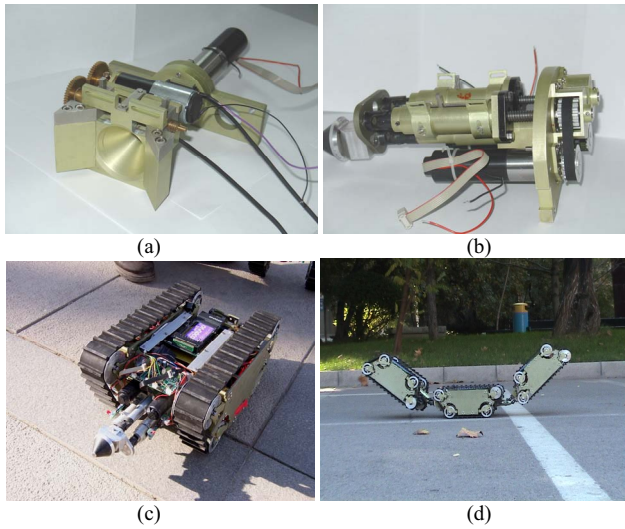


Fig. 5 Photos of the developed mechanisms, module and the robot. (a) The docking system and the serial part of the joint. (b) The parallel part of the joint. (c) A JL-I module. (d) The actual robot with three modules.

B. Electrical Design

The JL-I is a reconfigurable robot with many modules. Every module is an autonomous mobile robot capable of performing basic tasks such as reconnaissance and navigation. In addition, it is able to communicate with other modules and physically link to them to negotiate difficult environments. In order to achieve all these features, a distributed control system is needed, and every module processes the information and determines the motion by itself.

As illustrated in Fig. 6, the module has two level electronics. In the "low level electronics", motor control and power management are performed by AVR microprocessors.

They communicate over a RS485 bus to a single board computer in the "high level electronics" subsystem which does all of the path planning, motion generation and navigation functions. A 1 Kg Li-ion battery pack provides about 90Wh of energy, and power dissipation is about 13W.

The high level electronics has a powerful computer that communicates with other modules and the user by wireless Ethernet. It is equipped with a 400MHz Intel XScale PXA255 processor, 64MB of RAM, 4MB of Nor FLASH, 128M of Nand FLASH, USB master and slave interfaces, serial interface, I2C bus, PC104, PC104 plus interfaces, two CompactFlash slots. The computer interfaces to a wireless LAN card, infrared and sonar proximity sensors, a digital compass, a gyro and a GPS receiver with up to 20 cm precision in carrier phase differential mode (Novatel). All functionalities of the electronics inside the modules have been tested and work well.

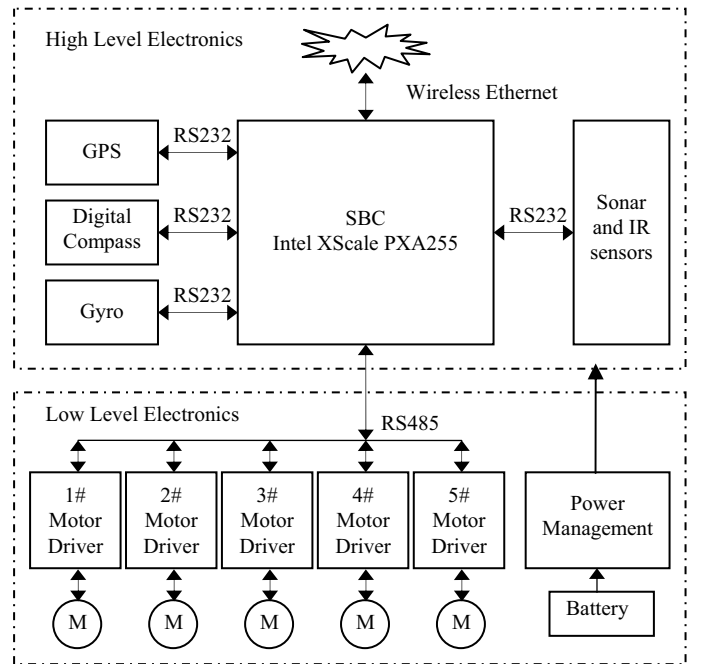


Fig.6 Electronics structure.

C. Software Architecture

The JL-I has many desirable properties including robustness and versatility. However, programming such a system with many modules is difficult and error-prone, as the configuration process always involves many modules in cooperation.

To cope with such difficult of programming, we proposed a software architecture for JL-I robot, which is based on a multiagent behavior-based structure and a multi-threaded programming methodology. The architecture as shown in Fig.7 includes a master serving as a global path planner and several slaves that just execute commands from the master. Each module could be a master or a slave depends on the requirements of the tasks. For handling of multiple hardware requests and computation in real-time, we build multiple tasks

on the processor, wherein the *ExDataCon* thread provides data communication with other modules and users, the *InDataCon* thread takes charge of receiving data or sending commands to the inner-devices, the *SenFuse* and *PathPlaning* thread process the sensor data, construct a local map and generate a feasible path, the *CompMotion* thread deals with composite behaviors composed by several primitive motions such as forward and turn around, which are generated by the *PrimMotion* thread. The communication threads (*ExDataCon* and *InDataCon*) maintain a robot status database that records all the modules' information. Thus, the configurations needed cooperation can be implemented more efficiently.

Once the global task commands are sent by the user, a module is chose to be a master module, and then the master module generate a set of composite motions on the basis of the task and all the modules' status. Afterwards, every composite motion is transmitted to the appropriate module by the *ExDataCon* thread. At last, the composite motions are decomposed and sent to the *PrimMotion* thread to achieve all the desirable motions.

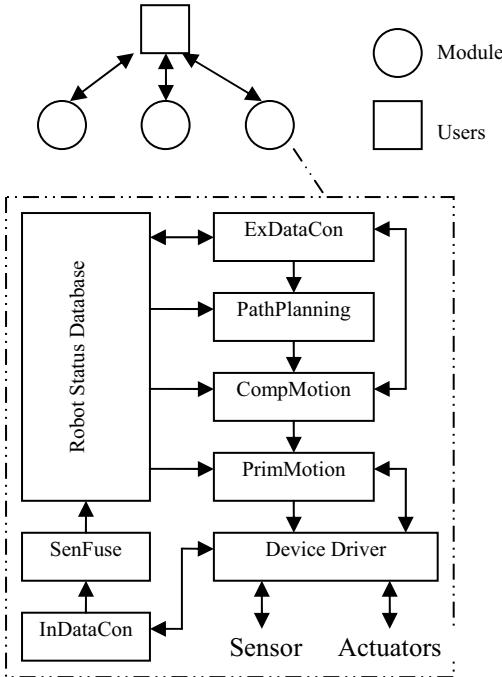


Fig. 7 Software architecture.

IV. MOTIONS OF RECONFIGURATIONS AND FUNCTIONALITIES

As described above, reconfiguration is achieved by repeating basic operations such as detaching a module from the robot, reconnecting the module to another module and changing the pose relationship. Fig.8 gives a diagram on how to implement these basic motions in the JL-I. The actual configuration process needs to combine these motions. As the number of the modules increases, these combined motions lead to a large variety of robot configurations. By arranging these configurations in a proper order, the robot can

implement tasks like getting over a step, self-righting and passing over a narrow space, which is difficult for a fixed-shape robot. In this section, 3 modules are sued to show the robot's functionalities.

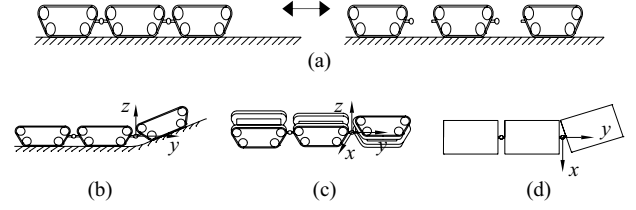


Fig.8 Basic motions of the JL-I robot. (a) Using docking systems to form a train configuration or separate the robot to several smaller robots. (b) Rotation around the x axis. (c) Rotation around the y axis. (d) Rotation around the z axis.

A. Getting over a step

The robot's configuration sequence to getting over a step nearing double height of the robot is shown in Fig. 9.

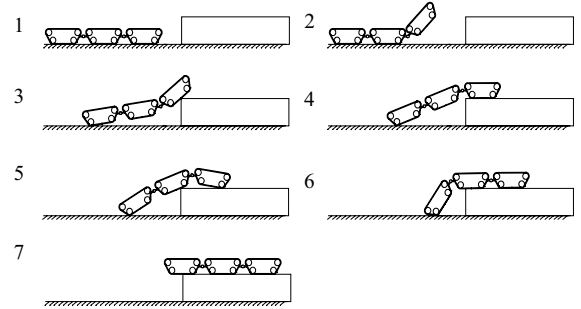


Fig. 9 Sequence of Getting over a step.

B. Self-righting process

The 3-DOF active spherical joints afford capabilities for 90°/180° self-righting. The configuration sequence is shown in Fig. 10.

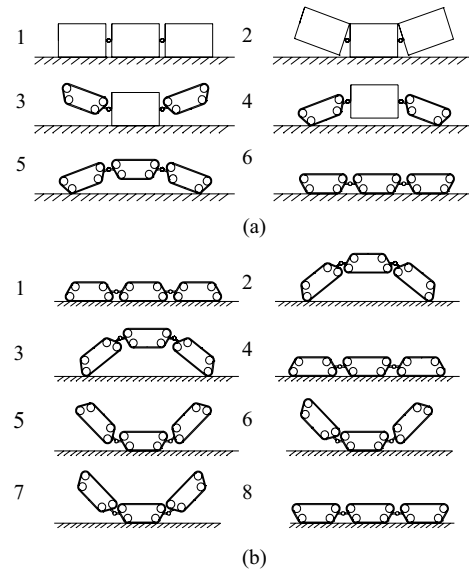


Fig. 10 Self-righting process. (a) Sequence of 90° self-righting process. (b) Sequence of 180° self-righting process.

C. Passing over a narrow space

As shown in Fig. 11, the train configuration robot is also able to pass over a fence narrower than the width of modules.

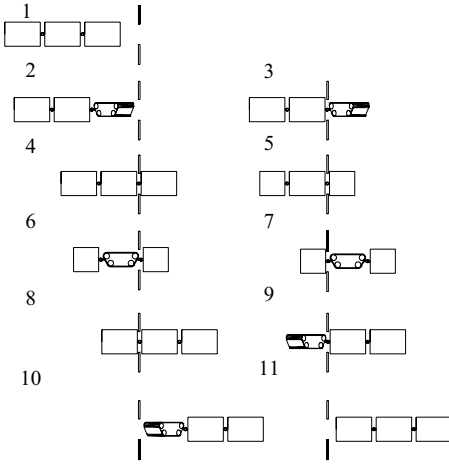


Fig. 11 Sequence of passing over a fence (Top View).

V. EXPERIMENTS

An actual robotic system with three modules was built and a series of experiments as shown in below have been performed to demonstrate the robot's motions of reconfigurations and functionalities discussed above.

A. Basic Motions of Reconfigurations

The 3-DOF active joints afford 3 basic motions. They are pitch, yaw and roll movement between two neighboring modules (Fig. 12).

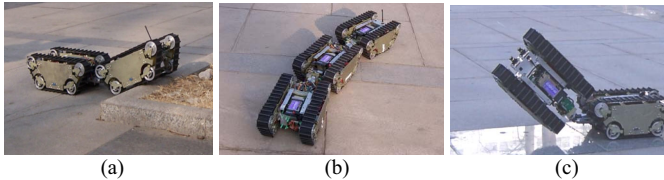


Fig. 12. Basic motions of the joints. (a) Pitch movement of the module. (b) Yaw movement of the module. (c) Roll movement of the module.

Fig. 13 shows the docking process of the connection mechanism, whose most distinctive features are its abilities to self-align and its high load strength. With the help of the powered tracks, the cone-shaped connector and the matching coupler are able to make the modules self-align $\pm 30\text{mm}$ lateral offsets and $\pm 45^\circ$ directional offsets.



Fig. 13. Sequence of docking process.

The experimental result shows that the active spherical joints and the docking mechanisms have the capability to implement the proposed configurations that provide the robot with self-righting capability and enhanced maneuverability for passing over obstacles. The specifications of the robot are given in Table I.

TABLE I
SPECIFICATIONS OF THE JL-I

Posture adjustment angle	
Rotation about X-axis	$-45^\circ \sim 45^\circ$
Rotation about Y-axis	$-45^\circ \sim 45^\circ$
Rotation about Z-axis	$0 \sim 360^\circ$
Allowable Misalignment (Docking mechanism)	
Lateral offset	$\pm 30\text{mm}$
Directional offset	$\pm 45^\circ$
Terrain adapting ability	
Maximum step height	280mm
Maximum ditch length	0~500mm
Minimum fence width	200mm
Maximum slope angle	40°
Self-righting ability	$0 \sim 180^\circ$
Module Size	
Length x Width x Height	430mmx250mmx155mm
Module Weight	Approx. 7Kg

B. Motions of the functionalities

The JL-I improves on the flexibility and adaptability by using novel active spherical joints between modules. We verify that the robot can implement all the motion functionalities described in section IV. Fig. 14~16 show the experimental results.

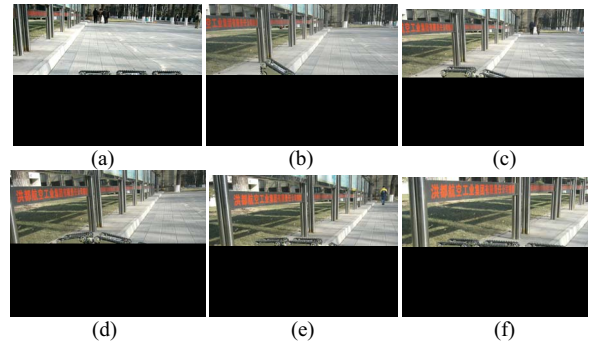


Fig. 14 Video sequence of getting over a step

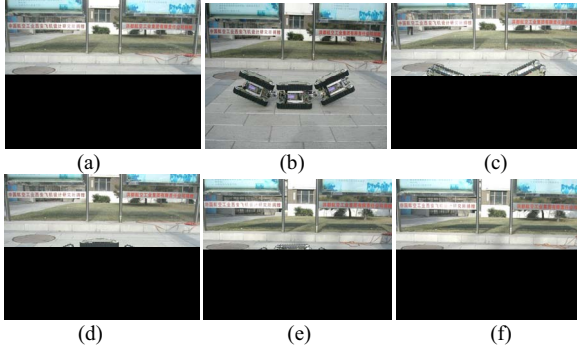


Fig. 15 Video sequence of the 90° self-righting.

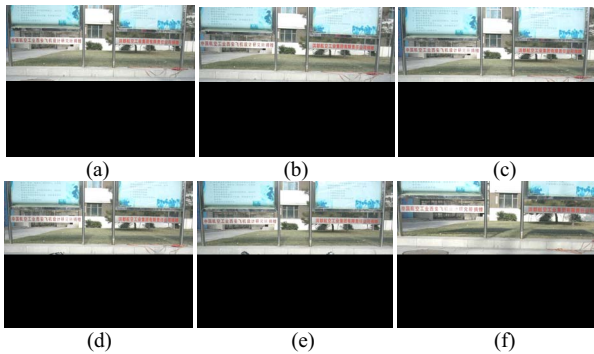


Fig. 16 Video sequence of the 180° self-righting.

VI. CONCLUSION

In this paper, we developed a new modular reconfigurable robot called JL-I. It consists of several modules which could disconnect from the robot to perform tasks in parallel or link into a train configuration for rough terrain. The hardware system and software architecture for the robot module were described and its basic reconfiguration and many functionalities were demonstrated through experimental results.

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